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Sustainable Mobility

Chapter 2 – Quantifying Transportation impacts





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STRUCTURE

- Modeling of traffic, energy and environmental performance of road vehicles
- Experimental Monitoring of traffic, energy and environmental performance of road vehicles

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Gasoline or Diesel Engine?

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EMISSIONS: GASOLINE VS DIESEL

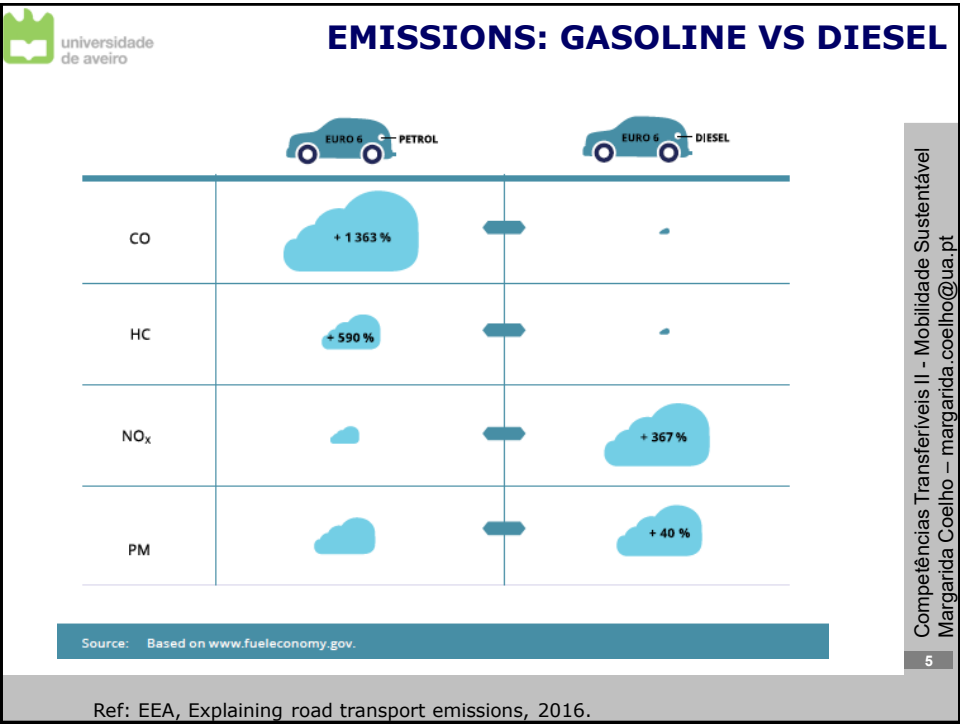
	HC	CO	NO _x	Particulate Matter
Diesel				
Gasoline				

- CO₂ emissions per liter: gasoline ~ Diesel
- Emissions per km: Diesel has lower consumption ⇒ Less CO₂ emissions

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TYPES OF EMISSIONS

Vehicles emissions can be categorised into 3 groups:

- **Exhaust emissions:** produced primarily from the combustion of different petroleum products such as petrol, diesel, natural gas (NG) and liquefied petroleum gas (LPG).
- **Abrasion emissions:** emissions produced from the mechanical abrasion and corrosion of vehicle parts. Abrasion is only important for PM emissions and emissions of some heavy metals. Significant levels of PM emissions can be generated from the mechanical abrasion of the vehicle's tyres, brakes and clutch, the road surface wear or the corrosion of the chassis, bodywork and other vehicle components.
- **Evaporative emissions:** the result of vapours escaping from the vehicle's fuel system. Evaporative emissions are important for only VOCs. Petrol fuel vapour contains a variety of different HCs, which can be emitted any time there is fuel in the tank, even when the vehicle is parked with its engine turned off.

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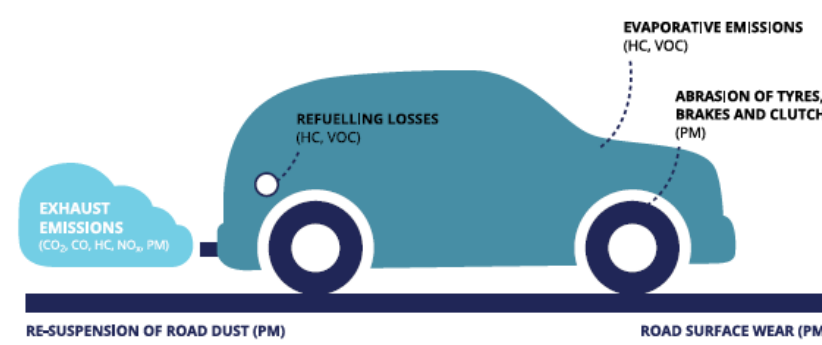
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TYPES OF EMISSIONS



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Ref: EEA, Explaining road transport emissions, 2016.

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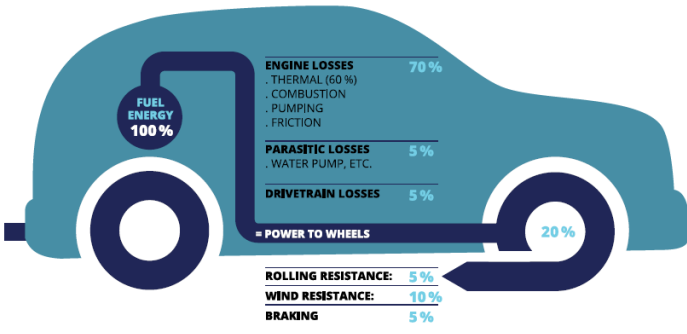


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VEHICLE EMISSIONS AND EFFICIENCY

Vehicle emissions and efficiency

In a conventional vehicle, only about 18 to 25 % of the energy available from the fuel is used to move it on the road, depending on the driving conditions. The rest of the energy is lost to engine and drivetrain ⁽¹⁾ inefficiencies. A small proportion of the energy produced is used to power vehicle accessories (e.g. radio, air conditioning). Therefore, the potential to further improve fuel efficiency using advanced technologies remains significant. While newer diesel engines remain more fuel efficient than petrol engines, their impact on air pollution is worse because of the higher levels of NO_x and PM that they emit.



Category	Percentage
FUEL ENERGY	100 %
ENGINE LOSSES	70 %
- THERMAL (60 %)	
- COMBUSTION	
- PUMPING	
- FRICTION	
PARASITIC LOSSES	5 %
- WATER PUMP, ETC.	
DRIVETRAIN LOSSES	5 %
POWER TO WHEELS	20 %
ROLLING RESISTANCE	5 %
WIND RESISTANCE	10 %
BRAKING	5 %

(¹) The drivetrain of a motor vehicle is the group of components that deliver power to the driving wheels. This includes the transmission, the axles and the wheels.

Ref: EEA, Explaining road transport emissions, 2016.

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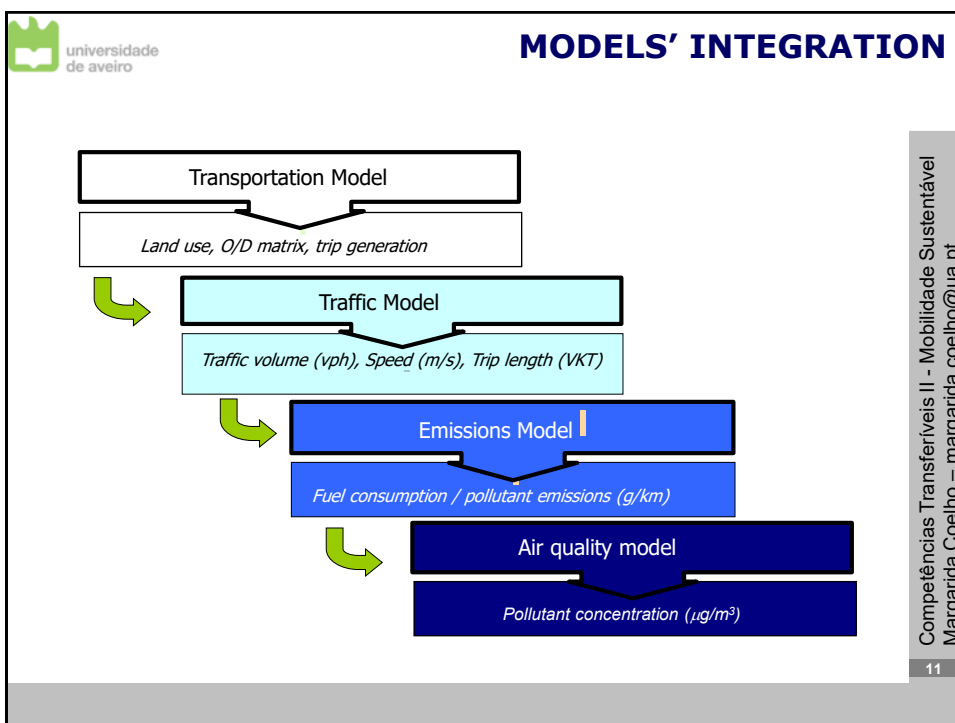


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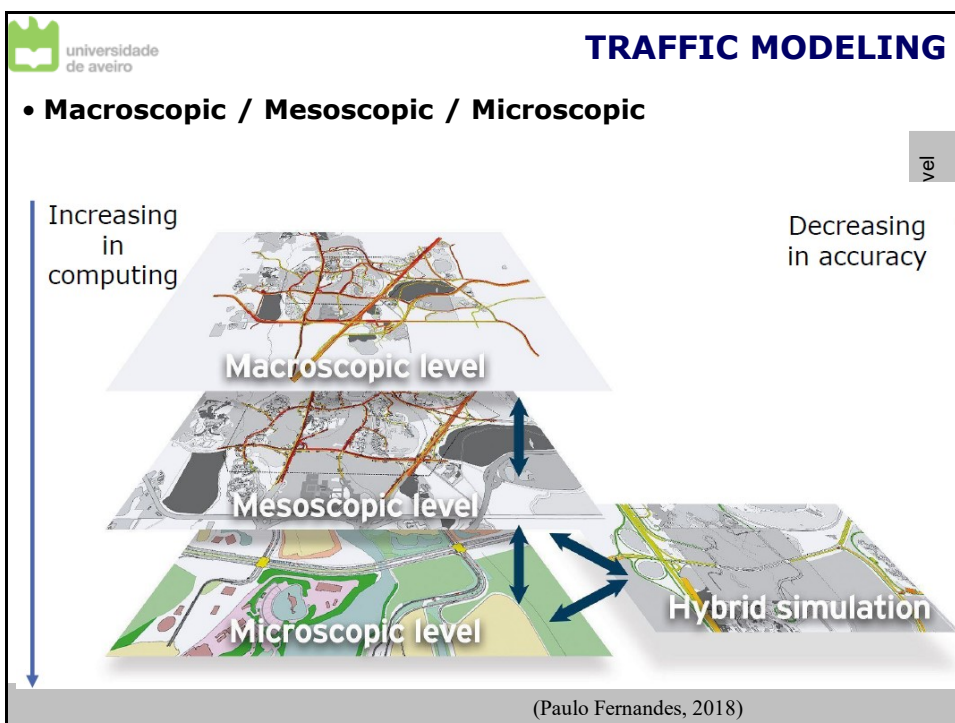
MODELS

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TRAFFIC MODELING

1. Simulation Traffic Models: Macroscopic

- Traffic is considered as a continuous flow
- Density and speed variables reflect traffic conditions

Disadvantages:

- Individual characteristics of vehicles are not considered;
- Impact of intersections are negligible.

Advantages:

- Processing Time;
- Modeling of large scale networks.

(Paulo Fernandes, 2018)

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TRAFFIC MODELING

2. Simulation Traffic Models: Mesoscopic

Traffic is considered as a platoon of vehicles

Disadvantages:

- Driving behavior parameters are not considered;
- Calibration procedure.

Advantages:

- Flexibility in preparing input data for the simulation;
- No much time consuming;
- Analysis can be made for some network sections;
- Modeling of Large scale Networks.

(Paulo Fernandes, 2018)

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TRAFFIC MODELING

3. Simulation Traffic Models: Microscopic

Each entity (vehicles or persons) is simulated individually

Disadvantages:

- Time consuming;
- Data collection of some inputs.

Advantages:

- Different vehicles types and characteristics;
- Capability of modeling driving behavior patterns;
- Impact of traffic uninterrupted facilities;
- Modeling of specific geometrical elements.

(Paulo Fernandes, 2018)

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TRAFFIC MODELING

• Examples of simulations:








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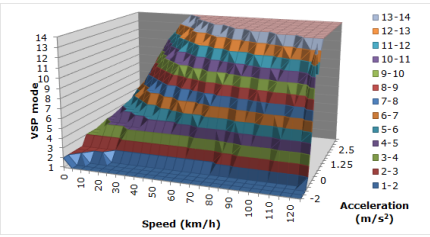
TRAFFIC TO EMISSIONS MODELING

Using traffic, fuel consumption/emissions, safety and noise models with an integrated approach

Reconstruction of vehicle trajectory



VSP



T. Fontes, P. Fernandes, H. Rodrigues, J. Bandeira, S.R. Pereira, A. Khattak & M.C. Coelho (2014) Are HOV/eco-lanes a sustainable option to reducing emissions in a medium-sized European city?, Transportation Research A: Policy and Practice, Vol. 63, May 2014, pp. 93-106.

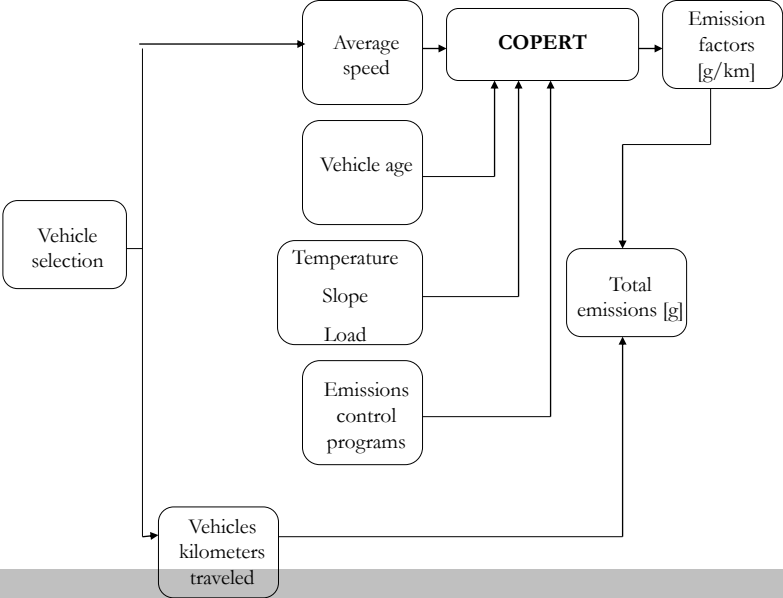
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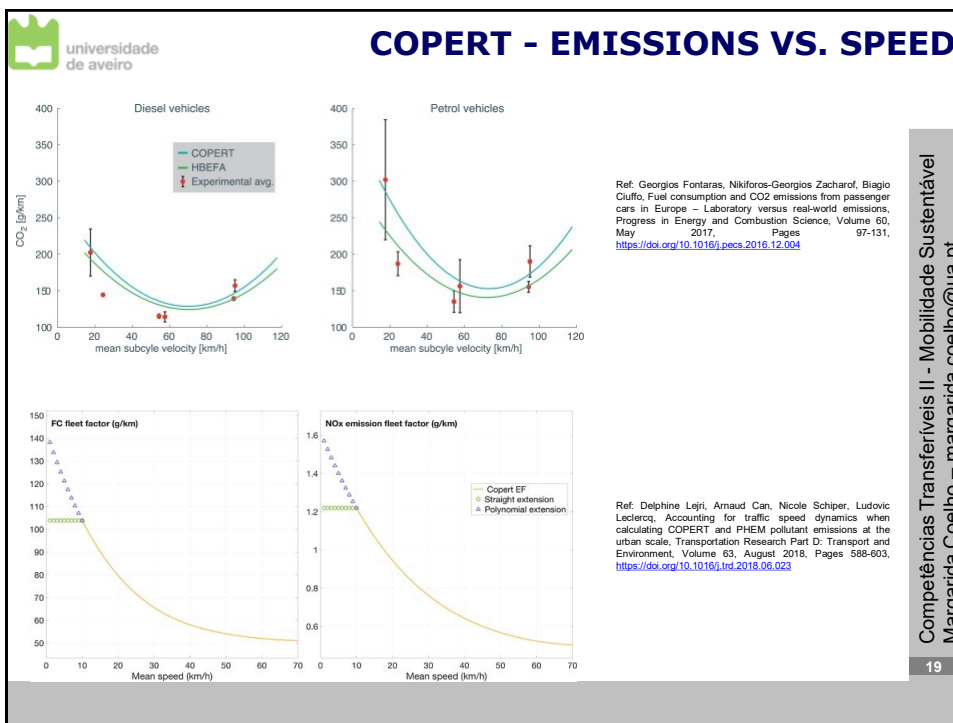
COPERT: METHODOLOGY



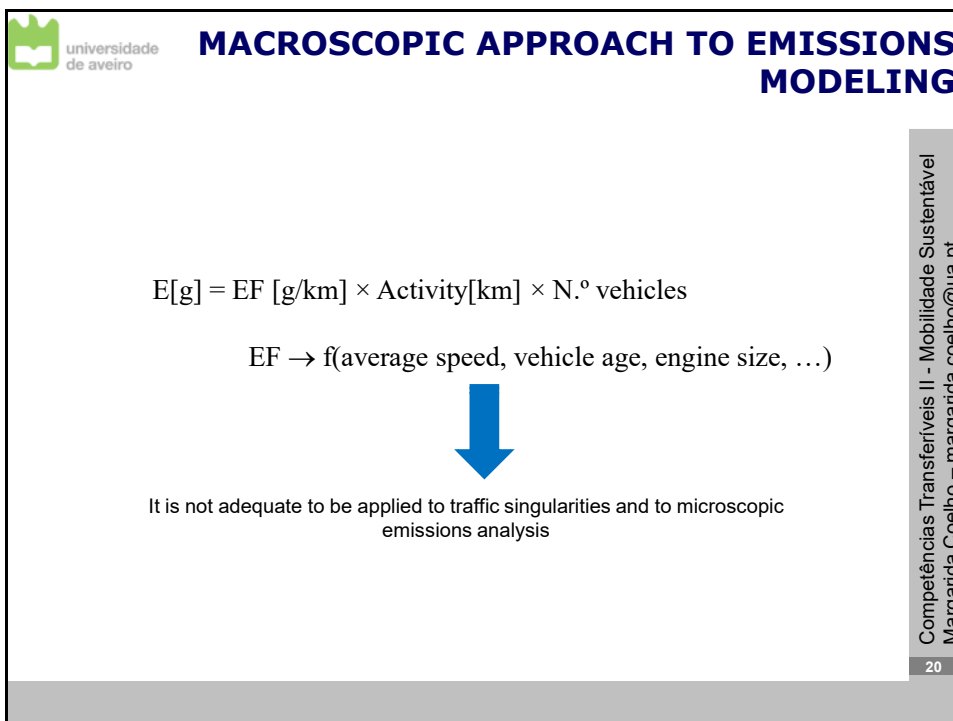
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MICROSCOPIC APPROACH TO EMISSIONS MODELING

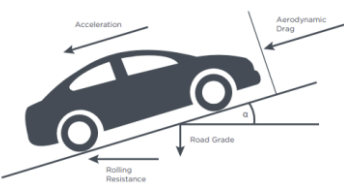
Vehicle Specific Power

$$\text{Vehicle Specific Power} = \frac{\frac{d}{dt}(KE + PE) + F_{\text{rolling}} \cdot v + F_{\text{Aerodynamic}} \cdot v}{m} =$$

$$= \frac{\frac{d}{dt} \left(\frac{1}{2} m \cdot (1 + \epsilon_i) \cdot v^2 + mgh \right) + C_R mg \cdot v + \frac{1}{2} \rho_a C_D A (v + v_w)^2 \cdot v}{m} =$$

$$= v \cdot (a \cdot (1 + \epsilon_i) + g \cdot \text{grade} + g \cdot C_R) + \frac{1}{2} \rho_a \frac{C_D \cdot A}{m} (v + v_w)^2 \cdot v$$

Ref: Jimenez-Palacios (1999) Understanding and quantifying motor vehicle emissions with vehicle specific power and TILDAS Remote Sensing, PhD Thesis, MIT.



Ref: ICCT (2016) Driving Resistances Of Light-Duty Vehicles In Europe: Present Situation, Trends, And Scenarios For 2025, White Paper.

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VSP ("VEHICLE SPECIFIC POWER")

Vehicle Specific Power

$$VSP = v \{ a(1 + \epsilon) + gr + gC_R \} + \frac{1}{2} \rho v^3 \left(\frac{C_D A}{m} \right)$$

Where

a	= vehicle acceleration (m/s ²)
A	= vehicle frontal area (m ²)
C_D	= aerodynamic drag coefficient (dimensionless)
C_R	= rolling resistance coefficient (dimensionless, ~ 0.0135)
g	= acceleration of gravity (9.8 m/s ²)
m	= vehicle mass (in metric tons)
r	= road grade
v	= vehicle speed (m/s)
VSP	= Vehicle Specific Power (kw/ton)
ϵ	= factor accounting for rotational masses (~ 0.1)
ρ	= ambient air density (1.207 kg/m ³ at 20 °C)

Frey, H.C., K. Zhang, and N.M. Roupail, "Vehicle-Specific Emissions Modeling Based Upon On-Road Measurements," *Environmental Science and Technology*, 44(9):3594-3600 (2010).

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VSP ("VEHICLE SPECIFIC POWER")

Vehicle Specific Power

Deceleration, or Downhill

Idle

Cruising, Acceleration, or Uphill

VSP Mode	Definition (kW/ton)
1	$VSP < -2$
2	$-2 \leq VSP < 0$
3	$0 \leq VSP < 1$
4	$1 \leq VSP < 4$
5	$4 \leq VSP < 7$
6	$7 \leq VSP < 10$
7	$10 \leq VSP < 13$
8	$13 \leq VSP < 16$
9	$16 \leq VSP < 19$
10	$19 \leq VSP < 23$
11	$23 \leq VSP < 28$
12	$28 \leq VSP < 33$
13	$33 \leq VSP < 39$
14	$39 \leq VSP$

Frey et al., Methodology for Developing Modal Emission Rates for EPA's Multi-Scale Motor Vehicle and Equipment Emission Estimation System, EPA420-R-02-027, 2002.

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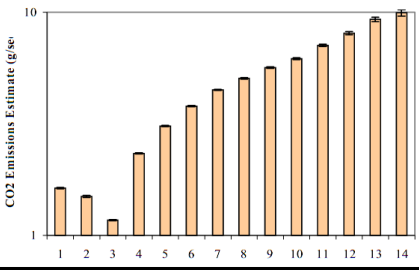
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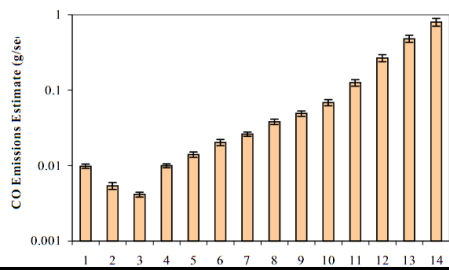
VSP ("VEHICLE SPECIFIC POWER")

Modo VSP	Definição
1	$VSP < -2$
2	$-2 \leq VSP < 0$
3	$0 \leq VSP < 1$
4	$1 \leq VSP < 4$
5	$4 \leq VSP < 7$
6	$7 \leq VSP < 10$
7	$10 \leq VSP < 13$
8	$13 \leq VSP < 16$
9	$16 \leq VSP < 19$
10	$19 \leq VSP < 23$
11	$23 \leq VSP < 28$
12	$28 \leq VSP < 33$
13	$33 \leq VSP < 39$
14	$39 \leq VSP$

CO2 Emissions Estimate (g/se)



CO Emissions Estimate (g/se)

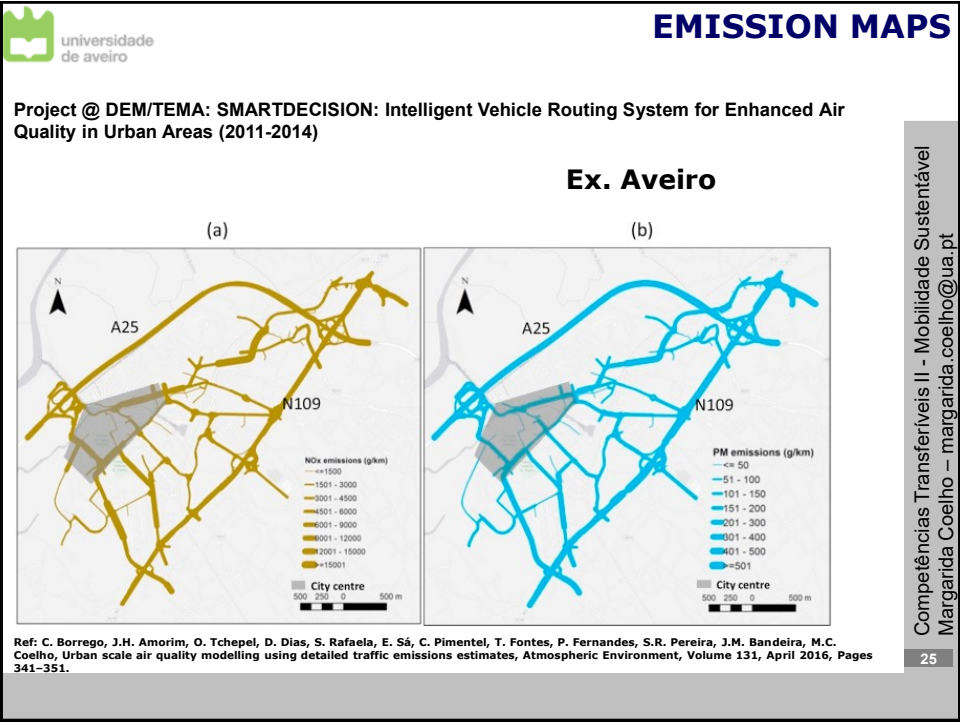


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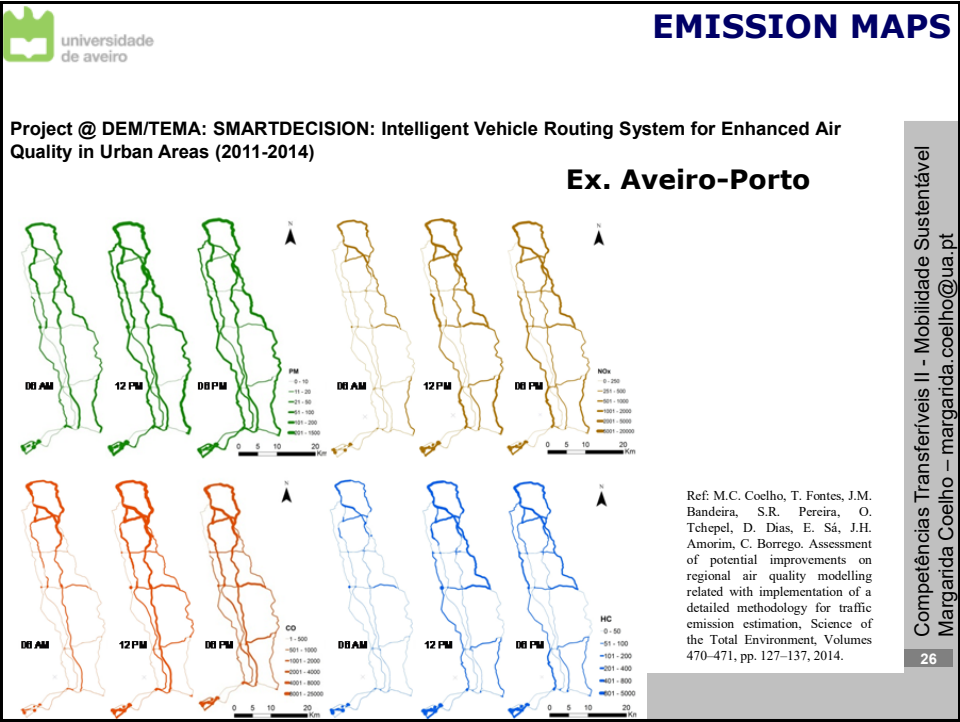
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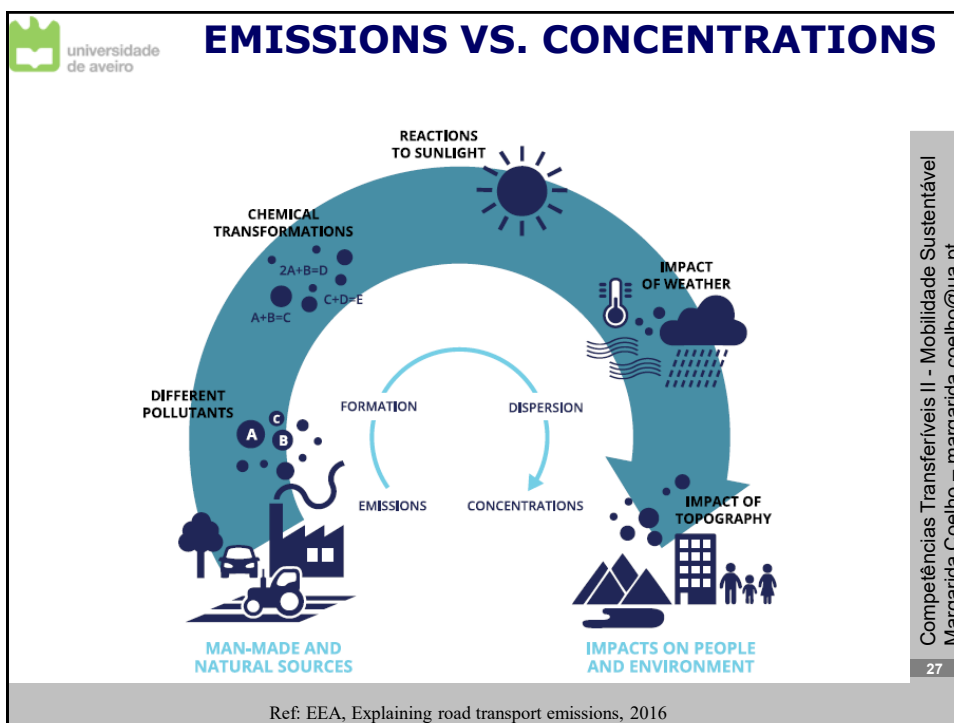
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POLLUTANT CONCENTRATIONS IN THE ATMOSPHERE

Combustion -> Emissions -> Dispersion

- **Gaussian models**
 - Used in the environmental impact procedures for mobile emission sources
 - **Base information:**
 - **Meteorology:** hourly registration of winds and temperatures, solar radiation, information about clouds
 - **Terrain and soil occupation:** 3D models of the terrain and soil occupation in order to define the obstacles to pollutant dispersion
 - **Emissions data:** emission rates for the studied pollutant, information about the emitting source (height, emission velocity, temperature)

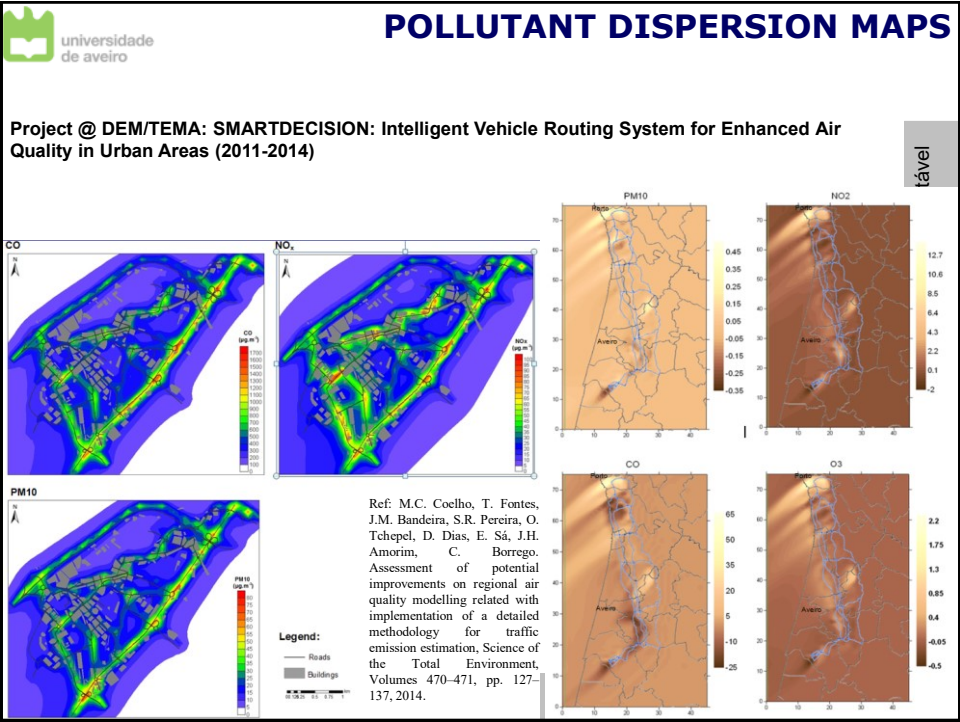
Some dispersion models: Caline 4, Highway II, Urban Airshed Model (UAM)

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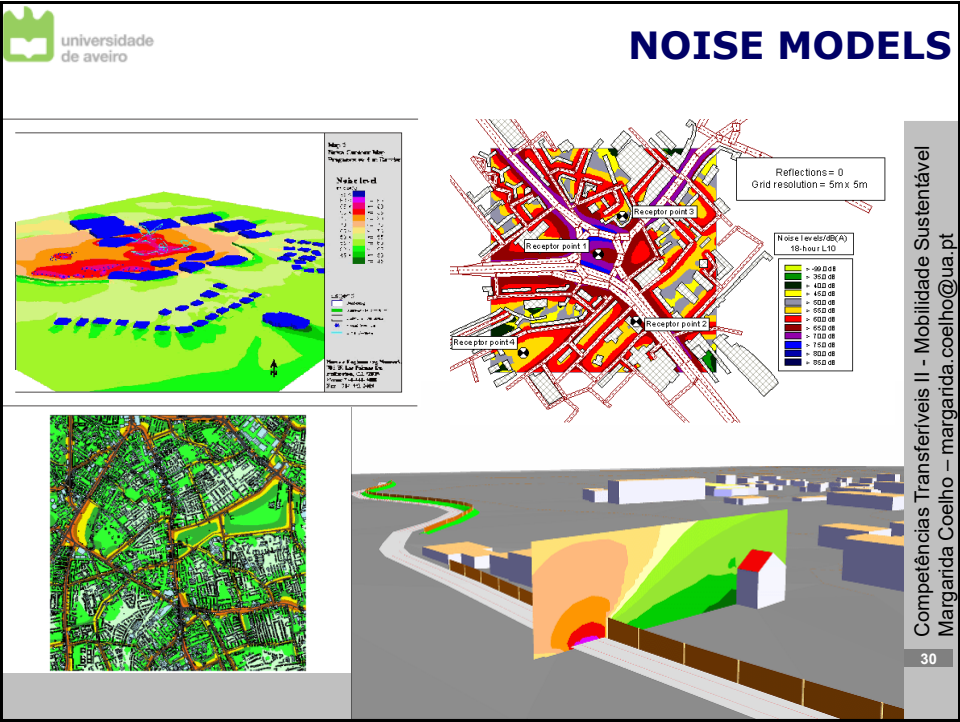
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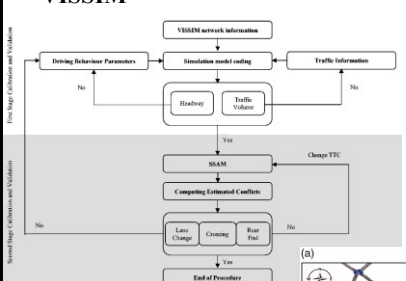
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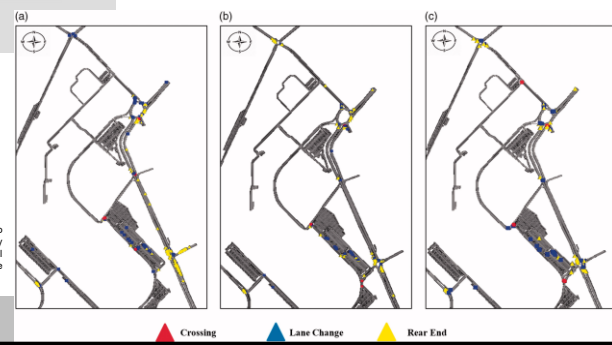
TRAFFIC TO SAFETY MODELING

Using traffic & safety models with an integrated approach

VISSIM



SSAM
Hotspot conflicts location



P. Fernandes, C. Sousa, J. Macodo & M.G. Coelho. How to evaluate the extent of mobility strategies in a university campus: An integrated analysis of impacts, International Journal of Sustainable Transportation, Volume 14, 2019 - Issue 2.

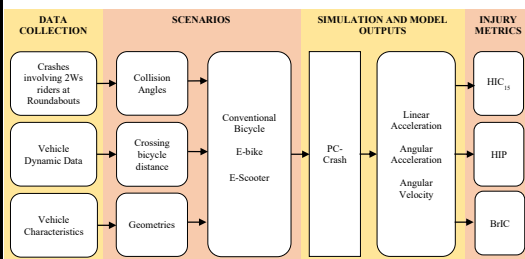
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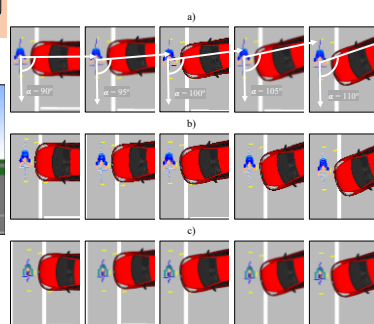
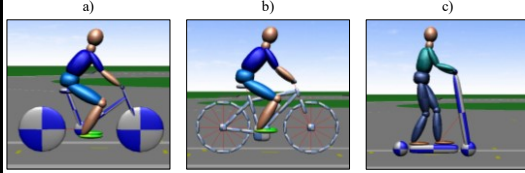


CRASHES RECONSTRUCTION

PC-CRASH model



HIC15 – Head Injury Criterion; HIP – Head Impact Power; BrIC- Brain Injury Criterion



Ref: Fernandes, Neves and Coelho (2022) Influence of bicycle crossing location at urban multi-lane roundabouts on potential injuries sustained to conventional bicycles, electric bicycles and e-scooter riders using PC-Crash reconstruction, To be presented at the 101th Transportation Research Board Annual Meeting.

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Experimental monitoring of traffic, energy and environmental performance

- Traffic
- Energy consumption & Pollutant emissions
- Pollutant dispersion
- Noise

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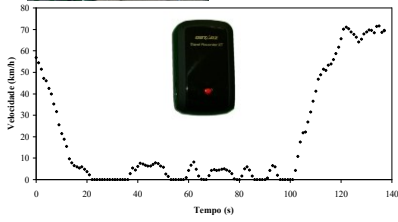
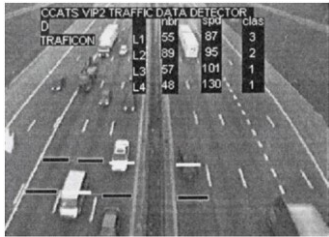
EXPERIMENTAL MEASUREMENTS OF TRAFFIC PARAMETERS

Caracterization of traffic volumes and
traffic performance:

Videotaping $\Rightarrow$ Queue length

N.º of stop&go situations

Distance and time of each
stop&go



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EXPERIMENTAL MEASUREMENTS OF TRAFFIC PARAMETERS

Measurement devices with Doppler effect:



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What is the procedure to measure vehicle emissions?



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EMISSION MEASUREMENTS

- **Standardised measurements in laboratories** are used to check that vehicles meet the official requirements for exhaust emissions.

However the **official procedures currently used in Europe are not representative of real driving conditions**. This problem has led to the development of new measurement procedures as well as portable emission measurement systems to obtain better information on real driving emissions.

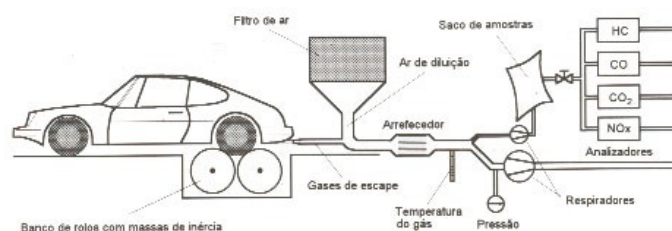
- According to Europe's laws, before being sold, vehicles must be tested to verify they are compliant with the required environmental, climate, safety and security standards. As it is not practical to test every single vehicle, one production vehicle is tested — considered representative of the 'type' — and, if all standards are respected, 'type approval' documentation is issued.

EMISSION MEASUREMENTS

- LDVs —passenger car, light commercial vehicle, moped or motorcycle — have to be tested on a '**chassis dynamometer**' or **roller bench**. A chassis dynamometer is designed to operate a vehicle indoors on a stationary platform to simulate real-world vehicle operation.

- The vehicle is driven on rollers, following a **predefined driving pattern**, with the dynamometer simulating the inertia of the vehicle, as well as the air drag resistance and the friction on the vehicle (known as the 'road load').

- The level of resistance on the dynamometer is adjusted for each specific vehicle tested to simulate the level of resistance that the vehicle would encounter if operated on the road, including: Vehicle aerodynamic resistance; Tyre rolling resistance.



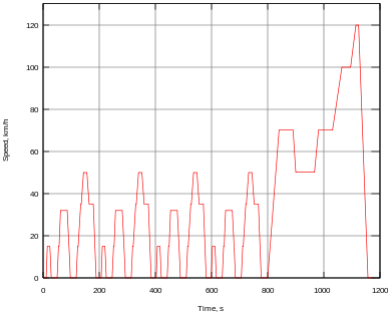


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EMISSION MEASUREMENTS

Previously... the so-called New European Driving Cycle (NEDC)

- First introduced in 1970 to represent typical driving conditions of busy European Cities; updated in 1990 in an attempt to better represent more demanding, high-speed driving modes.
- The NEDC now consists of an **urban and an extra-urban driving part.**
- The NEDC was **originally developed when vehicles were lighter and less powerful than those available today.** For these reasons, the test involves only a simple speed pattern with low accelerations, constant speed and many idling events that typically under-load modern day engines.
- It is widely accepted that the **NEDC is outdated**, with much evidence available from the scientific community and vehicle users clearly showing that the **emission values and fuel consumption measured in the laboratory largely understate the actual levels obtained under real-world driving conditions.** This difference occurs for a variety of reasons, including deficiencies of the NEDC testing procedure itself, but also due to certain deficits in the associated measurement protocols.



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THE CASE OF VW

Average emissions of nitrogen oxides in on-road testing

GRAMS OF NITROGEN OXIDES PER KILOMETER

Vehicle	Highway	Urban (Los Angeles)	Urban (San Diego)	Rural (Up and Downhill)
2011 Volkswagen Jetta	15 times limit	25 times	37 times	38 times
2012 Volkswagen Passat	9 times limit	20 times	17 times	17 times

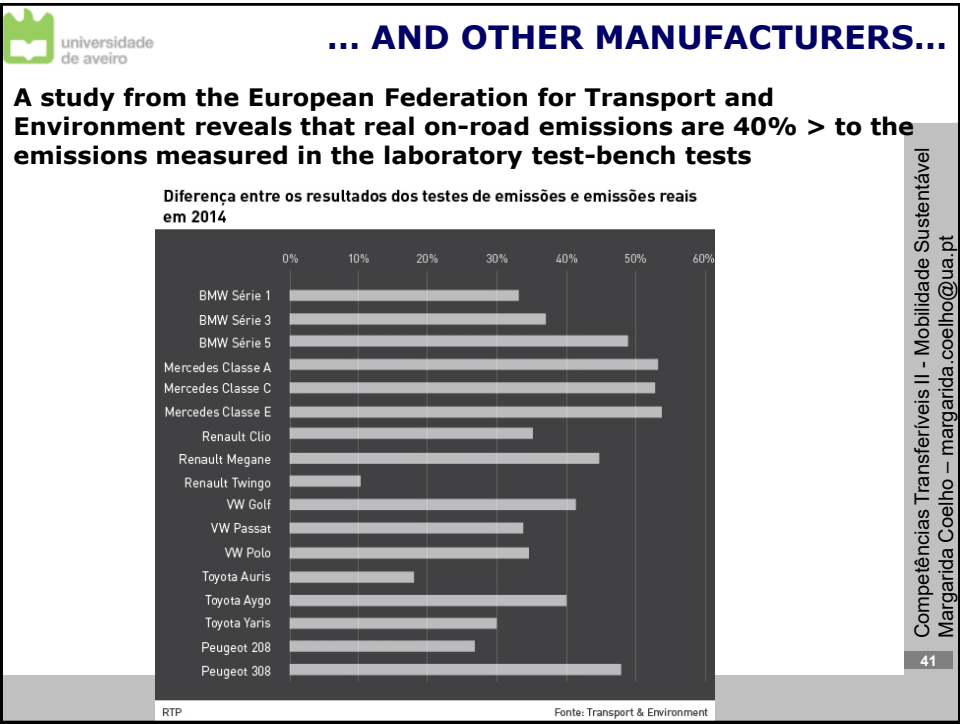
U.S. limit
.04 grams/kilometer

Source: Arvind Thiruvengadam, Center for Alternative Fuels, Engines and Emissions at West Virginia University

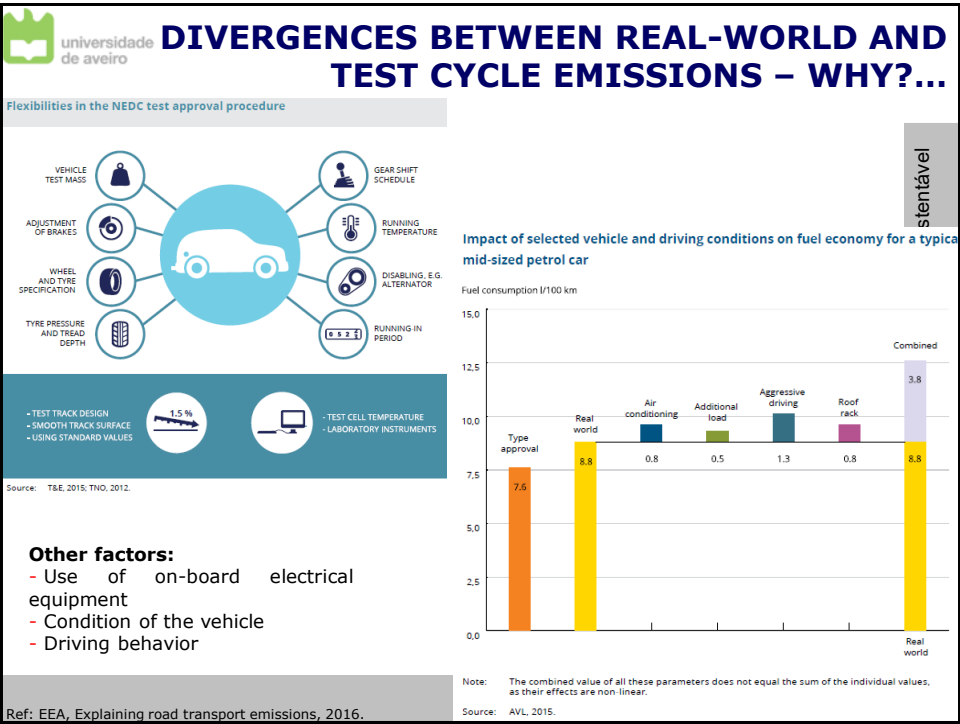
Fonte: <http://www.nytimes.com/interactive/2015/09/22/business/international/vw-volkswagen-emissions-explainer.html>

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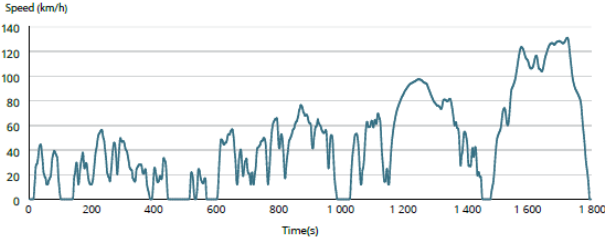
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WLTP TEST CYCLE

Speed profile of the WLTP test cycle



Source: GFEI, 2015.

Estimated impact of switching from the NEDC to the WLTP for an expected 2020 vehicle fleet

Regulatory Issue	NEDC	WLTP	Impact on CO ₂ emissions
Driving cycle	Operation at low loads with low engine efficiency, higher cold start effect (shorter distance), higher engine speeds (manual transmissions)	Higher speeds and acceleration forces, lower vehicle stop share (stop — start systems)	+ 2.1 %
Vehicle mass	No optional equipment No additional payload	Optional equipment: 70 kg Additional payload: 55 kg	+ 3.5 %
Temperature	Engine start temperature: 23 °C	Engine start temperature: 14 °C	+ 1.9 %
Total Impact 14 °C			+ 7.7 %
Total Impact 23 °C			+ 5.7 %

Ref: EEA, Explaining road transport emissions, 2016.

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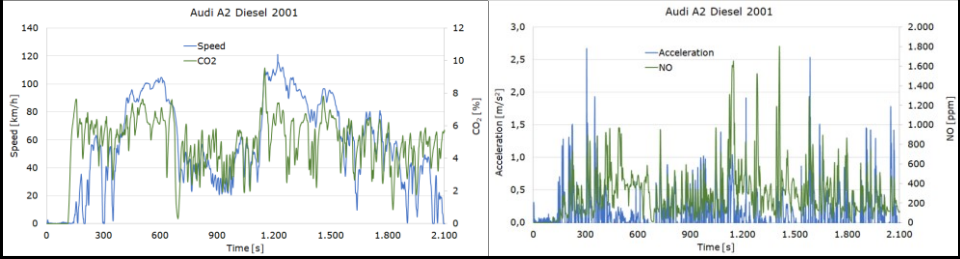
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EXPERIMENTAL MEASUREMENTS IN VEHICLES

Portable Emissions Monitoring System (PEMS)



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EXPERIMENTAL MEASUREMENTS IN VEHICLES

Portable Emission Monitoring Systems (PEMS) – Main limitations:

- **Reduced range of pollutants** that can be measured during a test compared with laboratory
- **Mass** (30–150 kg) it adds to the vehicle, which can affect the fuel consumption and hence measurements of the different pollutants.
- **Lower repeatability of measurements** encountered when testing, owing to real-world sources of variability, can be challenging to ensure consistency of measurements between different vehicles tested.

Advantages:

- Current laboratory emissions testing fails to capture the wide range of potential on-road emissions and that PEMS can assist in filling this gap.
- Past PEMS results show that average NO_x emissions of diesel cars, for the then-latest technology Euro 5 cars, substantially exceeded the Euro 5 emission limit by a factor of 4-7.
- On-road NO_x emissions of petrol vehicles, as well as CO and HC emissions of both diesel and petrol cars generally, stay within their emission limits.
- NO_x emissions were found to be the highest during uphill-downhill driving (rural) and during motorway driving at high speeds, i.e. at higher engine loads. This also provides an indication that the exhaust aftertreatment devices (the devices responsible for controlling exhaust air pollutant emissions) are underperforming under these operating conditions.
- CO₂ emissions tested with PEMS were also found to be higher (by 21 % on average) than laboratory tests for petrol and diesel cars. The magnitude of this discrepancy varies depending on vehicle type, operation mode, route characteristics and ambient conditions.

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AIR QUALITY STATIONS



Measurements of:

- Wind speed
- Wind direction
- Pollutant concentrations





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NOISE

- Some data...

1 heavy duty vehicle: 74 dB;

Noise for an avenue with a high traffic volume can reach 85dB

- Noise Law (Decreto-Lei n.º 278/2007, 1st August):

a) Sensitive zones cannot be exposed to a weighed continuous level of noise, LAeq, higher than 55 dB(A) in the day period and 45 dB(A) at night;

b) Mixed zones As zonas mistas cannot be exposed to a weighed continuous level of noise, LAeq, higher than 65 dB(A) in the day period and 55 dB(A) at night.

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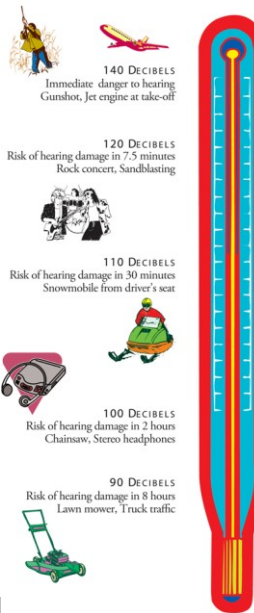
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MEASURING NOISE LEVELS

NOISE THERMOMETER



140 DECIBELS
Immediate danger to hearing
Gunshot, Jet engine at take-off

120 DECIBELS
Risk of hearing damage in 7.5 minutes
Rock concert, Sandblasting

110 DECIBELS
Risk of hearing damage in 30 minutes
Snowmobile from driver's seat

100 DECIBELS
Risk of hearing damage in 2 hours
Chainsaw, Stereo headphones

90 DECIBELS
Risk of hearing damage in 8 hours
Lawn mower, Truck traffic

125 DECIBELS
Pain threshold
Air raid siren, Firecracker

115 DECIBELS
Risk of hearing damage in 15 minutes
Baby's cry, Stadium football game

105 DECIBELS
Risk of hearing damage in 1 hour
Jackhammer, Helicopter

95 DECIBELS
Risk of hearing damage in 4 hours
Motorcycle, Power Saw

85 DECIBELS
Beginning of OSHA regulations

30 DECIBELS
Faint sound
Whisper



Sound level meter

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